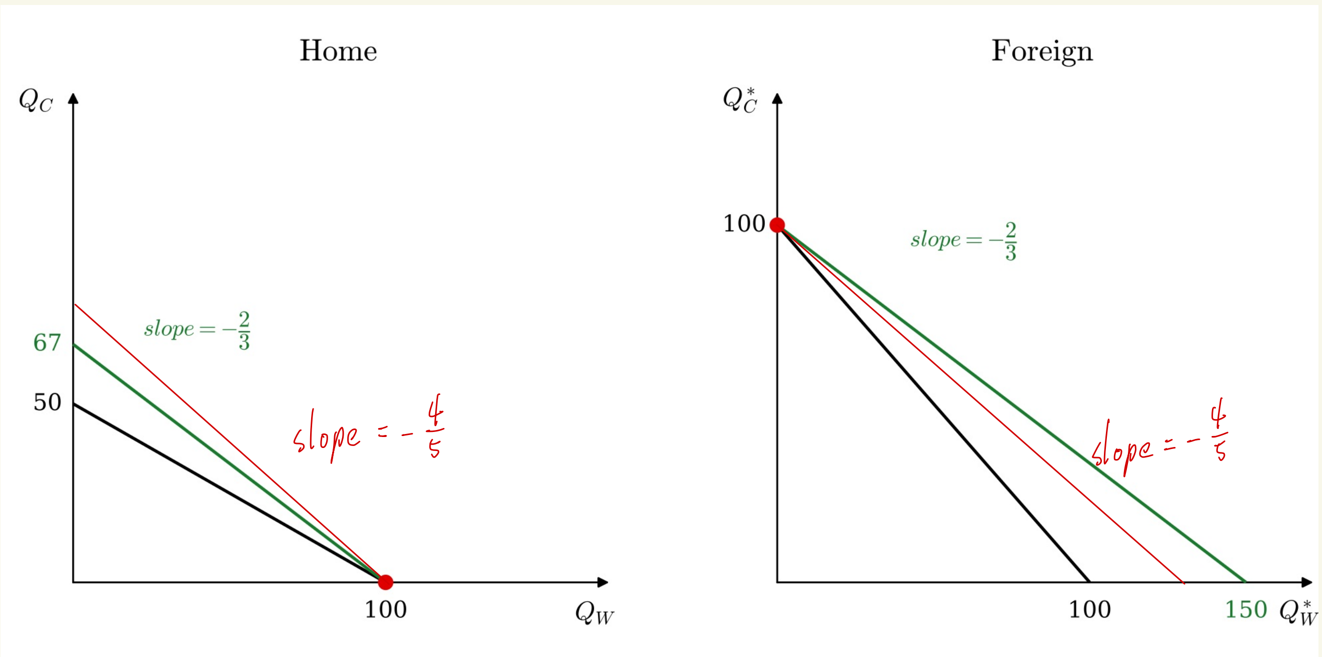
1. The black lines represent the countries' production possibilities frontiers (PPFs), which also define the consumption possibilities under autarky. Home has comparative advantage in wheat because its opportunity cost is ½ unit of cloth, compared with one unit in Foreign. Therefore, Foreign has a comparative advantage in cloth.

At the international relative price $P_W^I / P_C^I = 2/3$, Home specializes completely in wheat and Foreign specializes completely in cloth, as indicated by the red specialization points. The green lines are their consumption budget lines under trade, both with slope $-2/3$ and passing through their production points.

By exporting the good in which they have a comparative advantage and importing the other, both countries can consume combinations beyond their own PPFs and therefore gain from trade.

2. (a) $\frac{P_W^I}{P_W^C} = \frac{2}{3} \rightarrow \frac{4}{5}$



2 (b) Home is better off.

Home exports wheat. The increase in the price of wheat relative to cloth improves its terms of trade. As shown by the outward rotation of its budget line, Home's consumption possibilities expand.

Foreign is worse off

Foreign exports cloth and imports wheat. The world price change worsens its terms of trade. Its budget line rotates inward, reducing its consumption possibilities.

2(c) Yes, both countries are better off than under autarky.

The new international relative price remains between the two countries' autarky relative prices :

$$\frac{1}{2} < \frac{4}{5} < 1.$$

3.

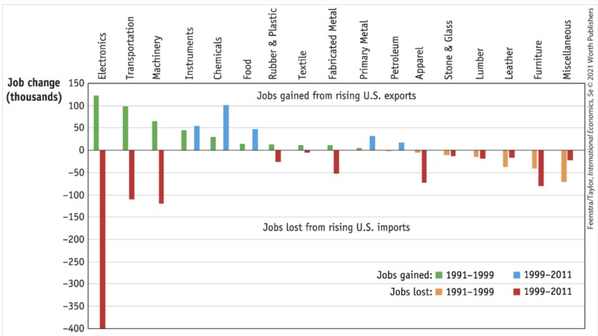


FIGURE 3-10 Job Gains and Job Losses in U.S. Manufacturing Industries, 1991–1999 and 1999–2011 For the period 1991–1999, estimated job gains due to rising U.S. exports are shown in green, and the estimated job losses due to rising U.S. imports are shown in orange. In this period, U.S. industries such as electronics, transportation, and machinery experienced the greatest job gains. For the period 1999–2011, estimated job gains due to rising U.S. exports are shown in blue, and the estimated job losses due to rising U.S. imports are shown in red. In this period, industries such as electronics, transportation, and machinery (which had experienced the greatest job gains over 1991–1999) now faced import competition, especially from China, and experienced job losses. Only a small number of industries such as chemicals, instruments, and food experienced job gains.

Data from: Robert C. Feenstra, Hong Ma, and Yuan Xu, 2019, "US Exports and Employment," Journal of International Economics, 120, 46–58.

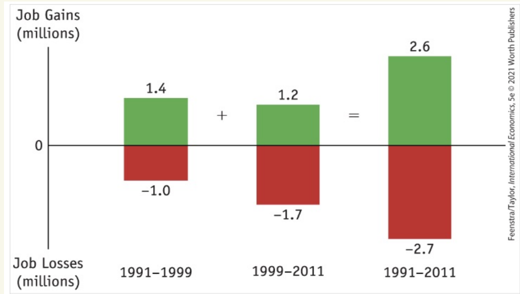


FIGURE 3-11 Total Job Gains and Job Losses in the U.S. Manufacturing Sector, 1991–2011 The estimated job gains due to rising exports are shown in green, and the estimated job losses due to rising U.S. imports are shown in red. These job gains and losses are the total for all manufacturing industries for the periods 1991–1999, 1999–2011, and 1991–2011. Job losses exceed job gains for the period 1999–2011 after China joined the WTO, but the losses are nearly balanced with the gains for the two decades 1991–2011.

Data from: Robert C. Feenstra, Hong Ma, and Yuan Xu, 2019, "US Exports and Employment," Journal of International Economics, 120, 46–58.

Over 1991 - 2011, rising US exports created approximately 2.6 million jobs, nearly offsetting the 2.7 million jobs lost because of rising imports, especially from China after 2001.

Export - related job (e.g., electronics, transportation, and machinery) gains exceeded import - related job losses in 1991 - 1999. However, the pattern reversed in 1999 - 2011.

Unlike the frictionless movement of labor assumed in the specific - factors model, it can take more than a decade for export job creation to balance the losses from a large import shock.

4.(a) $Q_M = 5K^{0.5}L_M^{0.5}$

$$MPL_M = \frac{\partial Q_M}{\partial L_M} = 2.5\left(\frac{K}{L_M}\right)^{0.5}$$

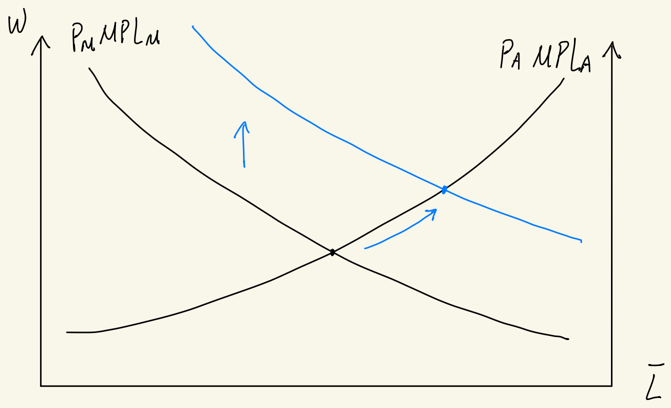
$$\Rightarrow P_M \times MPL_M = 2.5P_M\left(\frac{K}{L_M}\right)^{0.5}$$

① $K\uparrow \Rightarrow P_M \times MPL_M \uparrow$

② Productivity $\uparrow$ (e.g., $5 \to 6$) $\Rightarrow MPL_M' = \frac{\partial Q_M'}{\partial L_M} = 3P_M\left(\frac{K}{L_M}\right)^{0.5} > 2.5P_M\left(\frac{K}{L_M}\right)^{0.5}$

$\Rightarrow P_M \times MPL_M \uparrow$

(b) $K \uparrow \Rightarrow MPL_M = 2.5\left(\frac{K}{L_M}\right)^{0.5} \uparrow$



① equilibrium wage : $W = P_M MPL_M = 2.5 P_M \left(\frac{K}{L_M}\right)^{0.5} \uparrow$ & $L_M \uparrow$, $L_A \downarrow$

$\Rightarrow$ Mobile labor is better off because the wage rises in both sectors.

② Since P_M is fixed, $\frac{K}{L_M}$ should increase, implying an increase in K is more than an increase in L_M

The marginal product of capital : $MPK = \frac{\partial Q_M}{\partial K} = 2.5\left(\frac{L_M}{K}\right)^{0.5}$

Since $\frac{K}{L_M} \uparrow \Rightarrow \frac{L_M}{K} \downarrow \Rightarrow MPK \downarrow$

The rent for capital is $R_K = P_M MPK_M \downarrow$, so the owners of existing capital are worse off.

However, the total gains of K : $R_K \cdot K = 2.5 P_M \underset{(+)}{L_M^{0.5}} \underset{(+)}{K^{0.5}} \uparrow$ because $K \uparrow$ and $L_M \uparrow$

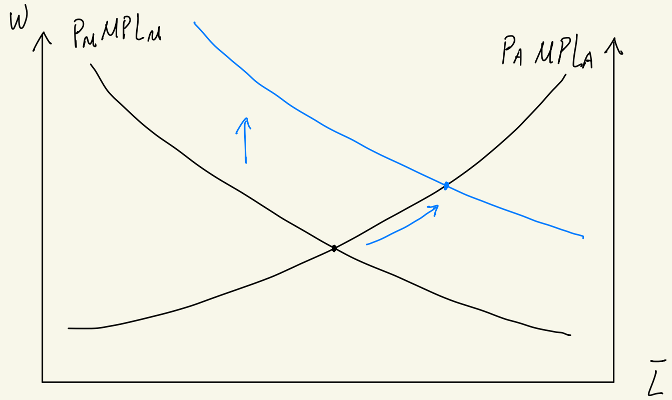
③ The marginal product of T : $MPT = \frac{\partial Q_A}{\partial T}$ is decreasing in T but increasing in L_A

T is fixed, but $L_A \downarrow \Rightarrow R_T = P_A MPT \downarrow$ & $R_T T \downarrow$ as well.

$\Rightarrow$ The owner of T is worse off.

(c) improve productivity : 5 → 6

$$MPL = 3\left(\frac{K}{L_M}\right)^{0.5}$$



① equilibrium wage : $W = P_M \times MPL_M \uparrow$ & $L_M \uparrow$, $L_A \downarrow$

⇒ Mobile labor is better off because the wage rises in both sectors.

② Since P_M and K are fixed, $\frac{K}{L_M}$ must decrease, implying the marginal product of K

$$MPK = \underset{\uparrow}{3}\underset{\uparrow}{\left(\frac{L_M}{K}\right)^{0.5}}$$ increases.

Therefore, the owner of K is better off

③ P_A & T are fixed, but $L_A \downarrow$ ⇒ $MPT \downarrow$

Hence, the owner of T is worse off

MPL ↑ due to	MPK	MPT	$\frac{W}{P_M}$ or $\frac{W}{P_A}$ ⇝ P_M & P_A are fixed
Increase in K	Falls (but K · MPK rises)	Falls	Rises
Increase in productivity in M	Rises	Falls	Rises